CCA Project

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February 4, 2026

Contents

1 Resultant

For the entire section, let $f, g \in \mathbb{K}[x]$, where $\mathbb{K}$ is a field, with $\deg f = n$ and $\deg g = m$.

First, we define the Sylvester matrix, $S(f,g) \in \mathbb{Z}^{(n+m)\times(n+m)}$, as follows [1] :

$$S(f,g) = \begin{pmatrix} f_n & & & & g_m & & \\ f_{n-1} & f_n & & & g_{m-1} & g_m & \\ \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots \\ f_0 & f_1 & \cdots & f_n & g_0 & g_1 & \cdots \\ & f_0 & \ddots & \vdots & & g_0 & \ddots \\ & & \ddots & f_1 & & & \ddots \\ & & & f_0 & & & g_0 \end{pmatrix}$$

where all the non-specified coefficients are null. We can observe that the first m columns of $S(f,g)$ are coefficients of $f(x)$ shifted m times, and the last n columns are coefficients of $g(x)$ shifted n times. We can also define the matrix with reverse order of f and g, or by shifting in rows instead of columns.

Then, the resultant of f and g is defined as [1]:

$$res(f,g) = \det S(f,g)$$

The resultant is important, as we have the following property:

$$res(f,g) \neq 0 \iff \gcd(f,g) = 1$$

which implies that the resultant of two polynomials is 0 if and only if they have a common root.

Proof. Let $\mathbb{K}[x]_{<d}$ be the $\mathbb{K}$-vector space with polynomials of degree $< d$, and let its standard bases $(1, x, \ldots, x^{d-1})$.

First, let us reexamine the definition of the Sylvester matrix: We define the $\mathbb{K}$ linear map:

$$\phi : \mathbb{K}[x]_{<m} \oplus \mathbb{K}[x]_{<n} \to \mathbb{K}[x]_{<n+m}$$

such that:

$$\phi(u,v) = uf + vg$$

This definition is a linear map since the addition of polynomials and multiplication of polynomials with a constant polynomial is linear.

Then, we can observe that the matrix of ϕ in the standard bases is exactly the Sylvester matrix, $S(f,g)$. Hence,

$$res(f,g) = 0 \iff \det(S(f,g)) = 0 \iff \det(\phi) = 0 \iff \ker(\phi) \neq \emptyset$$

As a result, we can write:

$$res(f,g) = 0 \iff \exists (u,v) \neq (0,0), \deg(u) < m, \deg(v) < n : uf + vg = 0$$

We show the contraposition, $\gcd(f,g) \neq 1 \iff res(f,g) = 0$.

($\implies$): Assume $\gcd(f,g) \neq 1$. Then $\exists h \in \mathbb{K}[x], \deg(h) > 0 : h \mid f$ and $h \mid g$. Let $f_1, g_1 \in \mathbb{K}[x] : f = hf_1, g = hg_1$ and set $u = g_1, v = -f_1$. Then,

$$uf + vg = g_1 h f_1 - f_1 h g_1 = h \times (g_1 f_1 - f_1 g_1) = 0$$

As $\deg(h) > 0$, we have $\deg(u) < m, \deg(v) < n$. Thus, by the above equality,

$$\gcd(f,g) \neq 1 \implies \exists (u,v) \neq (0,0), \deg(u) < m, \deg(v) < n : uf + vg = 0 \implies res(f,g) = 0$$

($\Longleftarrow$): Assume $res(f,g) = 0$. Then, by the above equality we have

$$(\exists (u,v) \neq (0,0), \deg(u) < m, \deg(v) < n : uf + vg = 0) \implies uf = -vg$$

We know $u \neq 0$, as g is not 0 and u and v cannot be both 0. Let $d = \gcd(u,g)$ such that:

$$u = du_1, g = dg_1 \text{ where } gcd(u_1, g_1) = 1$$

Putting this on the above equation:

$$(du_1) \times f = -v \times (dg_1) \implies u_1 f = -vg_1$$

$$\implies g_1 \mid u_1 f \implies g_1 \mid f \text{ since } gcd(u_1, g_1) = 1$$

Thus, we now have g_1 as the common factor of g and f. Now showing that g_1 is not constant. We know $\deg(u) < m = \deg(g)$:

$$\deg(u) < \deg(g) \implies \deg(du_1) < \deg(dg_1)$$

$$\implies \deg(d) + \deg(u_1) < \deg(d) + \deg(g_1) \implies \deg(u_1) < \deg(g_1)$$

Since we have $\deg(u_1) \geq 0$, $\deg(g_1) > 0$. This shows that g_1 is a non-constant common divisor of f and g, so $\gcd(f,g) \neq 1$.

We conclude that

$$res(f,g) \neq 0 \iff \gcd(f,g) = 1$$

$\square$

2 Subresultants

As before, for the entire section, let $f, g \in \mathbb{Z}[x]$ with $\deg f = n$ and $\deg g = m$. Moreover, we assume $m \leq n$.

Resultant of polynomials is important, as it gives as an efficient manner to understand if two polynomials have a common root or not. However, it does not give any other information but about gcd. Thus, we can extend the resultant definition into subresultants, which include all results of the Extended Euclidean Algorithm [2].

First, we define Sylvester matrices for subresultants, similar to the resultant's matrix. Given $0 \leq k \leq m$,

$$S_k(f,g) = \begin{pmatrix} f_n & & & & g_m & & & & \\ f_{n-1} & f_n & & & g_{m-1} & g_m & & & \\ \vdots & & \ddots & & \vdots & & \ddots & & \\ f_{n-m+k+1} & \cdots & \cdots & f_n & g_{k+1} & \cdots & \cdots & g_m & \\ \vdots & & & \vdots & \vdots & & & & \ddots \\ f_{k+1} & \cdots & \cdots & f_m & g_{m-n+k+1} & \cdots & \cdots & \cdots & \cdots & g_m \\ \vdots & & & \vdots & \vdots & & & & & \vdots \\ f_{2k-m+1} & \cdots & \cdots & f_k & g_{2k-n+1} & \cdots & \cdots & \cdots & \cdots & g_k \end{pmatrix}$$

, which can be understand as removing the last k-columns of each submatrix of f and g, and then removing the last $2k$ rows of the matrix.

Then, the k-th subresultant of f and g is defined as [2]:

$$\alpha_k = \det S_k(f,g)$$

3 Multivariate Polynomials

Let $n > 1$ and $f, g \in \mathbb{Z}[x_1, \ldots, x_n] = \mathbb{Z}[x_2, \ldots, x_n][x_1]$, with $\deg f = n$ and $\deg g = m$.. Then, by the definition of the resultant as the determinant of the Sylvester matrix we know that $Res_{x_1}(f, g) \in \mathbb{Z}[x_2, \ldots, x_n]$. Moreover, we can write that:

$$\deg(Res_{x_1}(f, g)) \leq \deg_{x_1}(g) \times n + \deg_{x_1}(f) \times m$$

Informally, this results can be deduced from the fact that determinant is defined as the sum over all permutations of n of multiplication of coefficients in the Sylvester matrix. The coefficients of Sylvester matrix are either 0, a coefficient of f, or g. Thus, we can upper bound the degree of each term with the degree of the multiplication on each term. Here, an important point is that we shift coefficients of f exactly m times while coefficients of g n times. Thus, terms belonging to f (this is $\deg(f) = n$) are multiplied with the degree of g over x_1, which is $\deg_{x_1}(g)$, and vice versa for g. Finally, we sum both degrees to express the upper bound of the multiplication of all terms, which is the degree of the biggest term on the left (depending on f) times the degree of the biggest term on the right (depending on g).

Then, let $\eta \in \mathbb{Z}^{n-1}$, $f(\eta), g(\eta) \in \mathbb{Z}[x_1]$. Then, we cannot say that the below holds for any chosen η:

$$Res_{x_1}(f(\eta), g(\eta)) = Res_{x_1}(f, g)(\eta)$$

In other terms, if $O \subseteq \mathbb{Z}^{n-1}$ such that $\forall \eta \in O : Res_{x_1}(f(\eta), g(\eta)) = Res_{x_1}(f, g)(\eta)$, then $O \neq \mathbb{Z}^{n-1}$.

The reason behind this is the elimination of variables in $\mathbb{Z}$: Since $\mathbb{Z}$ is a ring, but not a field, not every element has a multiplicative inverse and thus some elements may become 0 after specialization. This means that evaluating the polynomial on η may actually result in some coefficients dissapearing. If the dissapearing coefficient is the leading coefficient, this means that the degree of the polynomial changes, which changes the size of the matrix that we are taking the determinant of. Thus, evoluating the polynomial before taking the *det* may change the degree, and thus the degree of the final polynomial evoluated as well. However, if we evoluate after taking the *det* then the leading coefficient does not dissapear until the evoluation, thus the resulting degree may become actually higher.

Basically, the main problem is possible evoluations of polynomials that may change the degree of the polynomial and these are the only points that the equality is not preserved (which is logical since the linear mapping (i.e. Sylvester matrix) and the determinant are both linear operations, so the order of evoluation should not change anything). Moreover, if we define the exact same over a field but not a ring, such a $\mathbb{R}$, then the equation holds and $O = \mathbb{R}$

References

[1] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra.* Cambridge University Press, Cambridge, 1999. Section 6.3: The resultant.

[2] Joachim Von Zur Gathen and Jürger Gerhard. *Modern Computer Algebra.* Cambridge University Press, Cambridge, 1999. Section 6.10: Subresultants.